1. (a) I think we find that example is the path graph P_4 with vertices

$$1 - 2 - 3 - 4.$$

We observe this graph is connected. It has no Euler circuit because vertices 1 and 4 have odd degree:

$$\deg(1) = \deg(4) = 1.$$

Therefore, it has no Hamilton cycle because it has no cycle at all.

(b) An example is the graph made of two triangles sharing exactly one vertex. WLOG, we will let the vertices be v, a, b, c, d, with edges

$$va,\ ab,\ bv,\ vc,\ cd,\ dv.$$

This graph is connected. Its degrees are

$$\deg(v) = 4, \qquad \deg(a) = \deg(b) = \deg(c) = \deg(d) = 2,$$

so every vertex has even degree. Therefore it has an Euler circuit, for example

$$v - a - b - v - c - d - v.$$

It does not have a Hamilton cycle because v is a cut vertex. If v is removed, then we find the graph becomes disconnected, but any graph with a Hamilton cycle cannot have a cut vertex (as observed in lecture).

(c) An example is K_4. A Hamilton cycle is

$$1 - 2 - 3 - 4 - 1.$$

But each vertex of K_4 has degree 3, which is odd, so K_4 has no Euler circuit.

(d) An example is the cycle graph C_4:

$$1 - 2 - 3 - 4 - 1.$$

This is a Hamilton cycle since it visits every vertex exactly once before returning to the start. It is also an Euler circuit since it uses every edge exactly once.

2. We observe that in the complete graph K_n, every vertex has degree

$$n - 1.$$

A connected graph has an Euler circuit iff every vertex has even degree. Therefore, since K_n is connected for all $n \geq 2$, K_n has an Euler circuit when

$$n - 1 \text{ is even.}$$

This happens when n is odd. So K_n has an Euler circuit for

$$n = 3, 5, 7, \ldots$$

(and also $n = 1$ if the trivial graph is included).

A connected graph has an Euler trail but not an Euler circuit iff it has exactly two vertices of odd degree. In K_n, all vertices have the same degree $n - 1$, so either all vertices are even or all are odd. Therefore the only way to have exactly two odd vertices is when there are exactly two vertices, namely $n = 2$.

So K_n has an Euler trail but not an Euler circuit exactly when

$$n = 2.$$

3. Yes, the graph is planar.

Label the vertices as in the picture. The graph has all edges of K_6 except

$$ad, \quad bf, \quad ce.$$

So it is

$$K_6 - \{ad, bf, ce\},$$

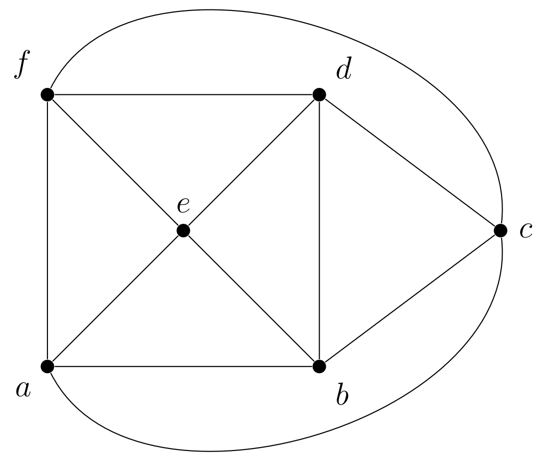
which is the same as

$$K_{2,2,2}$$

with parts

$$\{a, d\}, \qquad \{b, f\}, \qquad \{c, e\}.$$

A planar sketch is shown below.



This drawing has no crossings, so the graph is planar.

4. Since G is 4-regular,

$$\sum_{v \in V} \deg(v) = 4|V|.$$

Simply, we find

$$4|V| = 2|E| = 2 \cdot 16 = 32,$$

so

$$|V| = 8.$$

Now, we wish to use Euler's formula for a connected planar graph:

$$|V| - |E| + |F| = 2.$$

Substituting $|V| = 8$ and $|E| = 16$, we get

$$8 - 16 + |F| = 2,$$

so

$$|F| = 10.$$

Therefore the planar depiction has 10 regions.

5. (a) The edges of the given tree are

$$2-3,\ 3-4,\ 4-1,\ 1-5,\ 1-7,\ 6-7,\ 7-8.$$

Now repeatedly remove the smallest leaf and record its neighbor. And we perform the subsequent (repetitive) steps:

Initially the leaves are $2, 5, 6, 8$.

Remove 2, record 3.

Remove 3, record 4.

Remove 4, record 1.

Remove 5, record 1.

Remove 1, record 7.

Remove 6, record 7.

Therefore the Prüfer code is

$$(3, 4, 1, 1, 7, 7).$$

(b) We decode the Prüfer code

$$(5, 7, 2, 2, 3, 4).$$

The vertex set is $\{1, 2, 3, 4, 5, 6, 7, 8\}$.

At each step, connect the smallest label not appearing in the current code to the first entry of the code.

The code is $(5, 7, 2, 2, 3, 4)$. The smallest label not in it is 1, so add edge

$$1-5.$$

The remaining code is $(7, 2, 2, 3, 4)$. The smallest label not in it is 5, so add edge

$$5-7.$$

The remaining code is $(2, 2, 3, 4)$. The smallest label not in it is 6, so add edge

$$6-2.$$

The remaining code is $(2, 3, 4)$. The smallest label not in it is 7, so add edge

$$7-2.$$

The remaining code is $(3, 4)$. The smallest label not in it is 2, so add edge

$$2 - 3.$$

The remaining code is (4). The smallest label not in it is 3, so add edge

$$3 - 4.$$

The two vertices left at the end are 4 and 8, so add the final edge

$$4 - 8.$$

So the tree has edges

$$1 - 5,\ 5 - 7,\ 7 - 2,\ 2 - 3,\ 3 - 4,\ 4 - 8,\ 2 - 6.$$

A sketch of the tree is: